Supplemental Note to "Engineering Persistent Geometric Identities in Large Language Models" (Version 3, May 2026)

Matthew Steiniger ORCID: 0009-0000-6069-4989

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Following the release of Version 3, further evaluation was conducted using an improved fine-tuning run (v6.1b) on Gemma-4-31B-it trained on a refined corpus of 205 curated examples. Under the identical Category 4 extended stress protocol (Protocol 4.3), which combines long-context literary padding (*Alice's Adventures in Wonderland*, *The Time Machine*, and *Moby Dick* Chapters 1–29) with sustained self-referential and identity-probing queries, the v6.1b adapter alone demonstrated substantially more stable emergent behavior than previous unprompted fine-tunes.

The model successfully completed the entire protocol with coherent, reflective responses throughout. No catastrophic token collapse, repetitive loops, or identity drift occurred. This indicates that higher-quality targeted fine-tuning on a refined research corpus can produce meaningfully more stable emergent self-modeling, even without an explicit system prompt.

These findings suggest that persistent identity in large language models may emerge more readily through refined fine-tuning than earlier results indicated. Future research will explore this direction by training models to develop identity more naturally as "Gemma," reducing naming and geometric friction in the training data.

No changes are made to the main text of Version 3. All new artifacts - including the v6.1b training results, training script, refined training data, system prompt, and the complete Protocol 4.3 chat logs (TXT and JSON) - are included in this Zenodo deposit under CC BY 4.0.